Investigating Conductors & Insulators

Exam code: 0625 & 0972

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Electrical conductors & insulators

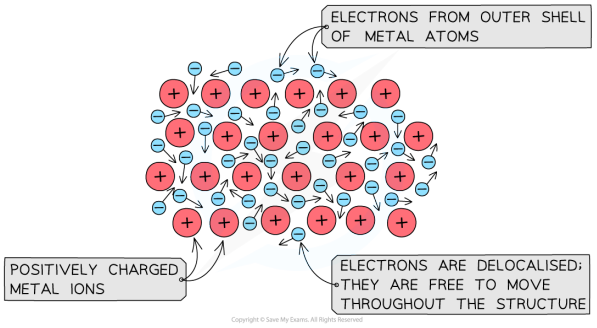
Conductors

- A **conductor** is a material that allows **charge** (usually electrons) to flow through it easily
- Some examples of conductors are:
  - silver
  - copper
  - aluminium
  - steel

- The best conductors tend to be **metals**
- On the atomic scale, metallic conductors are made up of positively charged metal ions with their outermost electrons **delocalised**
  - This means the electrons are free to move

- Metals conduct electricity very well because:
  - Current is the rate of flow of electrons
  - So, the more easily electrons are able to flow, the **better** the conductor

Metallic lattice structure



The lattice structure of a conductor with positive metal ions and delocalised electrons

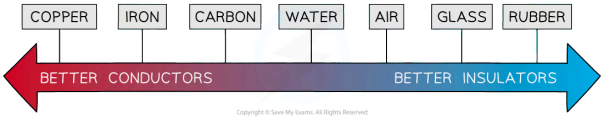
Insulators

Help me understand
Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- Some examples of insulators are:
  - rubber
  - plastic
  - glass
  - wood

- Some non-metals, such as wood, allow **some** charge to pass through them
- Although they are not very good at conducting, they do conduct a little in the form of **static** electricity
  - For example, two insulators can build up charge on their surfaces and if they touch this would allow that charge to be conducted away

Conductors and insulators



Different materials have different properties of conductivity

Investigating electrical conductors & insulators

Aim of the experiment

- The aim of this experiment is to distinguish between good and bad conductors using a gold leaf electroscope

Variables

- **Independent variable** = Different materials
- **Dependent variable** = Electrical conductivity of materials

Equipment

Equipment list

Equipment	Purpose
Gold-leaf electroscope	to distinguish between electrical conductors and insulators
Polythene rod	to charge the electroscope by contact
Cloth	to charge the rod by friction

plastic, glass, wood	electroscope when uncharged

Method

1. Before beginning the experiment, ensure the plate of the electroscope is uncharged by touching it with your finger. The leaf should hang straight down next to the stem
2. Charge a polythene rod by rubbing it with a cloth
3. Touch the charged rod to the plate of the electroscope. The leaf should stay risen if the electroscope has been successfully charged
4. Touch the plate of the charged electroscope with the first object to be tested and record any observations
5. Repeat for different materials

Example results table

material	gold leaf falls quickly or slowly	good or bad conductor
metal		
plastic		
glass		
graphite		
wood		
fabric		
rubber		

Analysis of results

- Good conductors, such as metals, allow charge to flow through them easily
  - Therefore, a good conductor will cause the leaf to fall **quickly** as it allows charge to flow to or from the plate
  - The **faster** the leaf falls, the **better** the conductor
- Poor conductors, such as glass, do not allow charge to flow as easily
  - Therefore, a poor conductor will cause the leaf to fall **slowly** as the charge is unable to flow as well
  - If the leaf does not move at all, the material is a **good insulator**
- The expected results are shown in the table below:

Table of conductors and insulators

metal	plastic
graphite	glass
	wood
	rubber
	fabric

Evaluating the experiment

- An electrometer (electronic instrument capable of measuring electric charge) could be used instead of an electroscope to allow for numerical comparisons between good and poor conductors

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